INCH-POUND

MIL-S-20708/21F
13 March 1989

SUPERSEDING
MIL-S-20708/21E
30 July 1987

MILITARY SPECIFICATION SHEET

SYNCHRO, CONTROL TRANSFORMER, TYPE 15CT6D

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Synchros described herein shall consist of this specification and the latest issue of MIL-S-20708.

TABLE I. Requirements

Requirement	Value	Unit	Tolerance
Frequency	60	Hz	±1%
Primary Voltage	78 (F)	volts	±1%
Primary Current	13.0	milliamps	maximum
Primary Power	0.25	watts	maximum
Impedance			
Zso	6000-7500	ohms	min-max
Zrs	1300-1650	ohms	min-max
Impedance Angle			
Zso	78.0-82.0	degrees	min-max
Zrs	13.0-18.0	degrees	min-max
Transformation Ratio	0.735	------	±2%
Phase Shift (Lead)	9.0	degrees	±2.0
Electrical Error	6.0	minutes	maximum
Null Voltage			
Total	65.0	millivolts	maximum
Fundamental	45.0	millivolts	maximum
Friction Torque	0.05	ounce-inches	maximum
Radial Play	0.0006	inches	maximum
End Play	0.0010	inches	±0.0005
Temperature Rise	10.0	degrees (C)	maximum

(F) denotes changes

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TABLE I Military part number variant characteristics

Military Part No 1/	A ± 010	B ± 010	G Maximum	Shaft Style	Terminal End	U 2/ + 0000 - 0002
M20708/21-01D	555	500	2 391	Splined	Terminal Board	1872
M20708/21-02D	↓	↓	2 391	Plain & Slotted	Terminal Board	↓
M20708/21-03D			2 129	Plain & Slotted	Wire Leads	

1/Military part number M20708/21-01D shall be the NATO standard.

1/The suffix letter following the numerical dash number corresponds to the latest modification letter of the type designation

(F) 2/The tolerance for plain and slotted shaft diameters is + 0000 and - 0005

TABLE III Military part number cross-references

SUPERSEDED MILITARY PART NUMBERS		NEW MILITARY PART NUMBE
MIL-S-20708/21D	MIL-S-20708/21C	
M20708/21D-001	M20708/21-001	M20708/21-01D
M20708/21D-002		M20708/21-02D
M20708/21D-003		M20708/21-03D

NOTE All line items shown in Table III refer to equivalent and interchangeable synchros of the same type designation modification Part number changes do not affect form, fit, or function of the synchros listed therein

Custodians
Army-AR
Navy-AS
Air Force-85

Review Activities
Army-MI,AV
Navy-OS
Air Force-11,17,99
DLA-ES

User Activities
Navy-MC,CG

Preparing Activity
Navy-AS

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